

Lab1: Gradient Descend

Nguyen Duc Duy - 2540043

May 6, 2026

This is my code for gradient descend with $f(x) = x^2$

```
def f(x):  
    return x*x  
  
def f_prime(x):  
    return 2*x  
  
def descend(x, alpha):  
    return x - alpha * f_prime(x)  
  
def gradient_descend(x, alpha, thred):  
    while abs(f_prime(x)) > thred:  
        print(x)  
        x = descend(x, alpha)  
    return x
```

Implementation steps:

1. Define the function $f = x^2$
2. Write the derivative function: $f' = 2 * x$
3. Function for compute the descend
4. The function gradient descend will do its job until x smaller than the thredshold.